Phys 20BL Week7/8 Out of Lab: RCL Circuits

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1 Physical Model for RC Circuit

1.1 Lab Guide Model

Let's start with what the lab guide says:

Given that the voltage of a capacitor is linearly proportional to the charge obtained on the capacaitor, if the proportionality constant is denoted as "capacitance" C, then it provides the following equation, with Q being the charge (unit: Coulomb C) and V being the voltage (unit: Volt V):

$$Q = CV \tag{1}$$

Also, by definition the rate of change of charge Q with respect to the time is the current of the system I (unit: Ampere A). Which, with capacitance C remaining constant, the above equation implies the following:

$$I(t) = \frac{dQ}{dt} = C\frac{dV}{dt} \tag{2}$$

Finally, using Ohm's Law, with $I = V/R$, where R is the resistance (unit: Ohms Ω). If the circuit has fixed resistance R, then one derives the following equation:

$$V(t) = I(t)R = RC\frac{dV}{dt} \tag{3}$$

Remark: This only considers the information across the capacitor, and fails to consider the general setup. Which, it can never solve the desired model on its own (as for $R, C > 0$ as assumption, this differential equation provides $V(t) = V_0 e^{t/(RC)}$, which is exponentially growing, failed to capture the general situation). So, let's include some more information.

1.2 Derivation for RC Circuits

Here, we'll fix a constant voltage source $V_s > 0$, $V(t)$ represents the voltage across the capacitor, $I(t)$ represents the total current in the circuit, C represents the constant capacitance of the capacitor, and R represents the constant resistance.

In an RC Circuit, since the voltage across the resistance and across the capacitance should sum up to the provided voltage V_0, hence one has voltage across the resistance being $V_0 - V(t)$. Also, by Ohm's Law, the voltage across the resistance is $RI(t)$, hence one has the equation $V_0 - V(t) = RI(t)$, or $RI(t) + V(t) = V_0$.

Finally, notice that $I(t) = \frac{dQ}{dt}$, where $Q(t)$ is the charge of the capacitor. Which, if $C, V(t)$ are the capacitance and voltage across the capacitor respectively, then $Q(t) = CV(t)$, hence $I(t) = \frac{dQ}{dt} = C\frac{dV}{dt}$.

Combine the two parts, we get the following differential equation:

$$RC\frac{dV}{dt} + V(t) = V_s \tag{4}$$

As a side note, for the case $V_s < 0$, this differential equation still works (just in general it's convenient to choose the direction where $V_s > 0$).

As of example, there are two functions (**Equation 1 & 2** in the In Lab portion) to consider:

$$V(t) = V_0(1 - e^{-t/\tau_{\text{step}}}) \tag{5}$$

$$V(t) = V_0^{-t/\tau_{\text{nat}}} \tag{6}$$

Let's try and solve these in terms of differential equations with initial conditions.

1.2.1 Solution for Equation 1

Given the equation $V(t) = V_0(1 - e^{-t/\tau_{\text{step}}})$, it models the circuit with the following condition:

- A fixed positive voltage source $V_s = V_0 > 0$.
- The capacitor is initially with no charge, namely $Q(0) = C \cdot V(0) = 0$. Then, it implies $V(0) = 0$.
- The resistance $R > 0$ and capacitance $C > 0$ of the circuit are fixed.

Which, solving the above differential equation under this condition, we get:

$$
\begin{aligned}
& RC\frac{dV}{dt} + V(t) = V_0 && (7)\\
\implies & \frac{dV}{dt} + \frac{1}{RC}V(t) = \frac{V_0}{RC} && (8)\\
\implies & \frac{dV}{dt}e^{t/(RC)} + V(t) \cdot \frac{1}{RC}e^{t/(RC)} = \frac{V_0}{RC}e^{t/(RC)} && (9)\\
\implies & \frac{d}{dt}\left(V(t) \cdot e^{t/(RC)}\right) = \frac{V_0}{RC}e^{t/(RC)} && (10)\\
\implies & V(t) \cdot e^{t/(RC)} = V_0 e^{t/(RC)} + D && (11)\\
\implies & V(t) = V_0 + De^{-t/(RC)} && (12)
\end{aligned}
$$

Under the initial condition $V(0) = 0$, one has $V_0 + D = 0$, or $D = -V_0$. Hence, the solution is given as follow:

$$V(t) = V_0\left(1 - e^{-t/(RC)}\right) \tag{13}$$

So, one has the step response time constant $\tau_{\text{step}} = RC$.

1.2.2 Solution for Equation 2

Given the equation $V(t) = V_0 e^{-t/\tau_{\text{nat}}}$. Notice that it's under the following condition:

- Zero voltage source $V_s = 0$ (modeling the situation where voltage source is turned off).
- The capacitor is fully charged initially, say $V(0) = V_0$ (normally meaning the voltage source before getting turned off).

Which, solving th above differential equation under this condition, we get:

$$
\begin{aligned}
& RC\frac{dV}{dt} + V(t) = 0 && (14)\\
\implies & \frac{dV}{dt} = -\frac{1}{RC}V(t) && (15)\\
\implies & V(t) = De^{-t/(RC)} && (16)
\end{aligned}
$$

Under the initial condition $V(t) = V_0$, oe has $D = V_0$. Hence, the solution is given as follow:

$$V(t) = V_0 e^{-t/(RC)} \tag{17}$$

So, one has the natural response time constant $\tau_{\text{nat}} = RC$ again.

2 Physical Model for LRC Circuit

This section is more related to the modification of experiments I've done.

Let's recall that for an inductor with fixed inductance L, when a current passes through it (assuming the current is a differentiable function), one has the following voltage change across the inductance:

$$V = -L\frac{dI}{dt} \tag{18}$$

Where, the voltage change is the voltage after the current passes through the inductor, subtracted by the voltage before the current passes through the inductor. In particular, its positive version $L\frac{dI}{dt}$ can be viewed as the voltage difference across the circuit.

2.1 Series and Parallel for Inductance

Suppose there are n inductors involved, with inductance $L_1, ..., L_n$ respectively. There are two cases to consider:

- If the inductors are in series, then notice they share a common current $I(t)$ (assume differentiability). Hence, across the ith inductor the voltage change is $V_i = -L_i\frac{dI}{dt}$, and the voltage change across the whole series arrangement is $V = \sum_{i=1}^n V_i = -\left(\sum_{i=1}^n L_i\right)\frac{dI}{dt}$.

  Which, if view the whole series arrangement with a total inductance L, then one has $V = -L\frac{dI}{dt}$, hence in general $L = \sum_{i=1}^n L_i$.

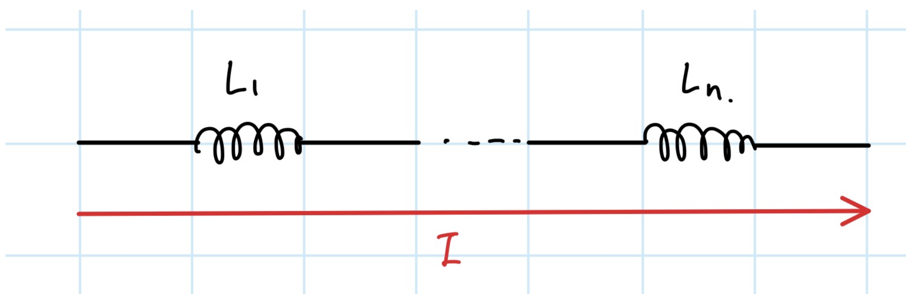

Figure 1: Inductors in Series

- If the inductors are in parallel, let I_i be the current passing through the ith inductor, then one has the total current $I = \sum_{i=1}^n I_i$. Then, notice they all share the same voltage change, say V. Hence, across the ith inductor one has voltage change $V = -L_i\frac{dI_i}{dt}$, or $-\frac{dI_i}{dt} = \frac{V}{L_i}$.

  As a result, the total current satisfies $-\frac{dI}{dt} = -\sum_{i=1}^n \frac{dI_i}{dt} = \sum_{i=1}^n \frac{V}{L_i}$, hence if L is a total inductance of the parallel arrangement, one has $V = -L\frac{dI}{dt} = V \cdot L\sum_{i=1}^n \frac{1}{L_i}$. In the general (nontrivial) situation, one has $\frac{1}{L} = \sum_{i=1}^n \frac{1}{L_i}$.

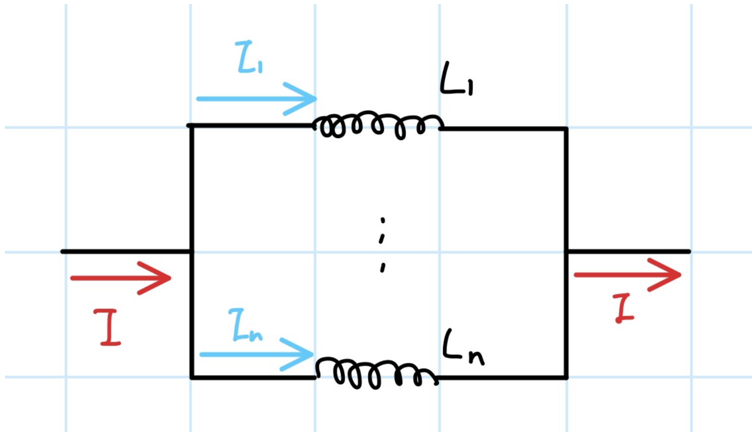

Figure 2: Inductors in Parallel

Hence, we concluded that under ideal situation, the current E&M Models predicted the total inductance to be:

- $L = \sum L_i$, given that the inductors are in series.
- $\frac{1}{L} = \sum \frac{1}{L_i}$, given that the inductors are in parallel.

2.2 Differential Equation for LRC Circuit

Now, let $V_s > 0$ represents a (fixed) voltage source, $V(t)$ be the voltage across the capacitance, L, R, C represents the inductance, resistance, and capacitance of the LRC circuit respectively. Then, with $I(t) = \frac{dQ}{dt} = C\frac{dV}{dt}$ be the current, one has $\frac{dI}{dt} = C\frac{d^2V}{dt^2}$.

Given the voltage across the inductor $L\frac{dI}{dt} = LC\frac{d^2V}{dt^2}$, the voltage across the resistance $RI(t) = RC\frac{dV}{dt}$, and the voltage across the capacitance $V(t)$, since they need to sum up to the fixed voltage source V_s, one arrives at the following differential equation:

$$LC\frac{d^2V}{dt^2} + RC\frac{dV}{dt} + V(t) = V_s \implies \frac{d^2V}{dt^2} + \frac{R}{L}\frac{dV}{dt} + \frac{1}{LC}V(t) = \frac{V_s}{LC} \tag{19}$$

Which, given the characteristic equation $r^2 + \frac{R}{L}r + \frac{1}{LC} = 0$, the roots are given by $r = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$ (which can be complex or real, dependent on the given value L). Which, labeling $w := \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}}$, the general solution to the homogeneous equation is:

$$V_h(t) = e^{-Rt/(2L)}\left(Ae^{wt} + Be^{-wt}\right) \tag{20}$$

Now, with the inhomogeneous part being constant $\frac{V_s}{LC}$, $V_p(t) = \frac{V_s}{LC}$ is a particular solution satisfying this differential equation, then the full solution is given as follow:

$$V(t) = V_s + e^{-Rt/(2L)}\left(Ae^{wt} + Be^{-wt}\right) \tag{21}$$

Finally, to obtain the voltage across the inductor, it's given as $LC\frac{d^2V}{dt^2}$, which can be solved once given all the coefficients.

2.3 Model for this Experiment

In our experiment, the provided voltage source is a square wave. However, realistically to observe the behavior of the LRC circuit, one only needs to observe the voltage within half a cycle, so one can treat the voltage source as 0 for $t < 0$, and $V_s = V_0$ for $t \geq 0$.

For simplicity of model, we'll also assume that within a short amount of time right after the voltage source is switched, the capacitance is not charged (which, it's approximated by $V(0) = 0$, or initially there's no voltage change across capacitance); on the other hand, one will also assume the voltage difference is given solely by the resistance, so $V_0 = RI(0) = RC\frac{dV}{dt}\Big|_{t=0}$. Hence, $\frac{dV}{dt}\Big|_{t=0} = \frac{V_0}{RC}$.

As for solution, notice that plugin to the general solution, one has the fllowing:

$$0 = V(0) = V_0 + A + B \implies A + B = -V_0 \tag{22}$$

$$\frac{V_0}{RC} = \frac{dV}{dt}\bigg|_{t=0} = -\frac{R}{2L}(A+B) + w(A-B) \implies A - B = \frac{1}{w}\left(\frac{1}{RC} - \frac{R}{2L}\right)V_0 \tag{23}$$

(Note: Here we assume $w \neq 0$, as for basically 100% of the L, R, C combinations, $w \neq 0$).

Hence, one obtains the following two coefficients:

$$A = \frac{1}{2w}\left(\frac{1}{RC} - \frac{R}{2L} - w\right)V_0, \quad B = -\frac{1}{2w}\left(\frac{1}{RC} - \frac{R}{2L} + w\right)V_0 \tag{24}$$

Which, if w is imaginary, the solution for V will be taken as the real portion.

3 Week 7 Portion - RC Circuit